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Roll No

CS-8203 (GS)
B.E. VIII Semester Examination, June 2020
Grading System (GS)
Data Mining and Knowledge Discovery
Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. Describe the differences between the following approaches for the integration of a data mining system with a database or data warehouse system: no coupling, loose coupling, semi-tight coupling and tight coupling. State which approach you think is the most popular and why.
2. Explain the following terms:
 - i) ROLAP
 - ii) MOLAP
 - iii) HOLAP
3. a) What is Data Warehouse? Explain the process to design data warehouse.
b) Define data marts and also discuss the types of data marts.
4. a) List out differences between classification and clustering methods with example.
b) What are the advantages of Decision Tree induction?
5. What are association rules? Define frequent Sets, Support and Confidence. Discuss importance of discovering association rules.
6. What is the “Apriori property”? How it is used by the Apriori Algorithm? Explain the limitation of Apriori Algorithm.

OR

Explain Pincer Search algorithm in detail. Demonstrate the working of this algorithm by taking suitable example dataset.

7. Apply Apriori algorithm for the finding frequent itemset for the following transaction with $\text{min_sub} = 3$.

TID	10	20	30	40
Item Brought	{A, C, D}	{B, C, E}	{A, B, C, E}	{B, E}

8. Write short notes on any two of the following:
 - i) Metadata Repository
 - ii) Confusion Matrix
 - iii) Data Cleaning
